

Tie Me Ghana SRS v2

Software Requirements Specification

Tie Me Ghana, A Hospital Communication System for Deaf and Hard of Hearing Patients

Version 2.0, ordered by feature priority Prepared for the MTN Ghana Tekyerema Pa Hackathon 2026

1. Introduction

1.1 Purpose

This document specifies the requirements for Tie Me Ghana, a lightweight installable web application enabling two way communication between hearing doctors or nurses and Deaf or Hard of Hearing (DHH) patients in Ghanaian hospitals. Requirements are organized by priority so the team builds the highest value, most feasible pieces first.

1.2 Scope

The system supports Twi as the spoken and written language, and Ghanaian Sign Language (GhSL) as the primary visual and sign channel, within a hospital consultation setting.

1.3 Definitions

Term	Definition
DHH	Deaf and Hard of Hearing
GhSL	Ghanaian Sign Language
ASR	Automatic Speech Recognition
TTS	Text to Speech
PWA	Progressive Web App
Gloss	The English word label used as the lookup key for a sign clip

2. Feature Priority Overview

Features are grouped into three tiers. P0 is the minimum needed for the system to function at all and is the primary hackathon build target. P1 features are strong differentiators that should follow once P0 is stable. P2 features are documented for direction but are explicitly out of scope for the current build.

P0, Core Consultation Flow (build first)

1. Doctor to patient captioning and GhSL sign video rendering
2. Literacy check and Guided Interrogation Mode for non literate patients
3. Patient to doctor response, spoken aloud in Twi or English
4. Session transcript logging

P1, High Value Additions (build once P0 works)

5. Emergency Visual Triage Mode
6. GhSL Prescription Playback with QR access

P2, Future Scope (not built now, documented for direction only)

7. Visual or vibration based queue call alternative
 8. Expansion to Ga and Ewe
 9. Extension beyond hospitals to lecture halls, churches, and public services
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3. Functional Requirements, by Priority

3.1 P0.1, Doctor to Patient Communication

FR 1.1 The doctor selects spoken input language, English or Twi, before speaking. **FR 1.2** Speech is transcribed using Automatic Speech Recognition. **FR 1.3** English text is translated to Twi text if needed. **FR 1.4** The Twi caption is displayed on screen. **FR 1.5** The caption is tokenized and matched to a library of pre recorded GhSL clips, using the English gloss as the lookup key. **FR 1.6** Unmatched words trigger fingerspelling fallback, playing individual letter clips in sequence. **FR 1.7** Matched clips are stitched into a single sign video and played alongside the caption.

3.2 P0.2, Literacy Check and Guided Interrogation Mode

FR 2.1 On first use, the system shows a sign video only prompt asking whether the patient can read and write, answered through two large icon based options with no text. **FR 2.2** The patient's answer is saved and determines their interaction path for the visit. **FR 2.3** Literate patients proceed to free captioning as described in 3.1 and to typed responses as described in 3.3. **FR 2.4** Non literate patients use Guided Interrogation Mode instead: the doctor selects a question from a preset clinical question bank, the app plays it as a GhSL sign video. **FR 2.5** For selection type questions, the patient taps from a grid of sign video answer options, for example body location. **FR 2.6** For yes or no questions, the app instructs the patient in sign video to nod or shake their head, observed directly by the doctor in person. No camera based gesture detection is used. **FR 2.7** The doctor confirms the observed answer by tapping Yes or No on their own screen, and this confirmation is what gets logged.

3.3 P0.3, Patient to Doctor Communication

FR 3.1 Literate patients type a response in English or Twi. **FR 3.2** Typed text is translated if needed and converted to speech. **FR 3.3** Non literate patients respond only through Guided Interrogation Mode as described in 3.2. **FR 3.4** Every patient response, typed or tapped, is spoken aloud to the hearing listener in either Twi or English, set once per session by the doctor or nurse. **FR 3.5** Spoken output occurs even for tap selected answers, so the response is audible without the doctor looking at the screen.

3.4 P0.4, Session Transcript

FR 4.1 Every exchange, in both directions, is logged with direction, text, and timestamp. **FR 4.2** The full transcript is stored only on the patient's own device by default, not uploaded to a shared server. **FR 4.3** The patient can view, scroll, and delete their own transcript.

3.5 P1.1, Emergency Visual Triage Mode

FR 5.1 The app provides one tap icons for a pain scale. **FR 5.2** The app provides a tappable body map for the patient to indicate pain location. **FR 5.3** The app provides pre recorded GhSL videos for critical alerts, including Asthma, Pregnancy, and Cannot Breathe. **FR 5.4** No typing is required anywhere in this mode. **FR 5.5** Every selection is spoken aloud in Twi or English, consistent with FR 3.4.

3.6 P1.2, GhSL Prescription Playback

FR 6.1 At the end of a consultation, the doctor's final instructions are saved as an ordered playlist of GhSL clips and Twi captions. **FR 6.2** The playlist is cached on the patient's own phone for offline replay. **FR 6.3** A QR code printed on the medicine envelope links to the same playlist through a small, de identified reference containing only the medicine name, dosage, and clip sequence. **FR 6.4** The full private consultation transcript is never included in this playlist and is not resolvable from the QR code.

4. User Interface Design Principles

Interaction design succeeds when a system is functional, intuitive, and pleasant to use. Every screen in Tie Me Ghana should be evaluated against five principles before it is considered finished. These are not abstract ideals, each one maps to a concrete requirement below.

4.1 Visibility

Relevant controls must be immediately visible, never hidden behind a menu the patient would need to already understand text or icons to find.

Application in this app:

- The language toggle (Twi or English output) and the literate or non literate path indicator are always visible on screen, never buried in settings.
- During Doctor to Patient communication, the caption and the sign video are shown together at all times, not as tabs the user must switch between.
- During an active recording or translation, a visible status indicator, for example a pulsing dot or waveform, shows the system is working, so the user is never left wondering if their action registered.

4.2 Feedback

Every user action must produce an immediate, unambiguous response, confirming the system received the input. Feedback in this app is deliberately both visual and physical, since a Deaf or Hard of Hearing user cannot rely on audio cues the way a hearing user would.

Application in this app:

- Visual feedback: a checkmark or highlight when a tap selection, for example a body part or a yes or no answer, is registered.
- Visual feedback: an animated waveform while spoken audio plays for the hearing listener, transitioning to a completed state, for example a checkmark, when playback ends.

- Vibration feedback: a short vibration pulse fires the moment a tap selection is registered, so the patient feels confirmation even if their eyes are on the doctor rather than the screen.
- Vibration feedback: a distinct vibration pattern fires when spoken audio to the hearing listener begins, and a different pattern when it finishes, giving the Deaf patient a physical cue for an event they cannot hear.
- Vibration feedback: in Emergency Visual Triage Mode, a stronger, more urgent vibration pattern confirms a critical alert selection, for example Cannot Breathe, reflecting the higher stakes of that action.
- Every vibration pattern used in the app must be distinct and consistent, so the patient learns to recognize what each one means over repeated use, this connects directly to the Consistency principle below.

4.3 Constraints

The interface should make incorrect or unsafe paths difficult or impossible, rather than relying on the user to avoid them.

Application in this app:

- A non literate patient is never shown a free text input field. Guided Interrogation Mode presents only tap based, pre validated options, structurally preventing a situation the patient cannot handle.
- The doctor's clinical question bank is a fixed, pre reviewed list, not an open text field, preventing an unreviewed or inaccurate sign video from ever being generated on the fly.
- The GhSL Prescription Playback QR reference can only resolve medicine name, dosage, and clip sequence. It has no technical path to expose the private consultation transcript, the constraint is built into what the reference contains, not left to a permissions setting that could be misconfigured.

4.4 Consistency

The same pattern, icon, and behavior should mean the same thing everywhere in the app, reducing what the user has to learn or remember.

Application in this app:

- The same tap to select interaction pattern is used across Guided Interrogation Mode and Emergency Visual Triage Mode. A patient who learns one already understands the other.
- The same icon set represents Yes and No everywhere in the app, in Guided Interrogation answers, in Emergency Triage, and in any future confirmation screen.
- The same video player appearance and controls are used for every sign video shown, whether it is a doctor's question, a patient's answer option, or a prescription instruction.
- Vibration patterns, as described in 4.2, are never reused for two different meanings, so a patient can trust that a given pulse always means the same thing.

4.5 Affordances

Elements should visually suggest how they are used, without requiring an instruction or a label to explain it.

Application in this app:

- The body map looks like an actual outline of a human body, so tapping the stomach or the head feels like pointing, not like operating a menu.
- Play controls on every video clip use a universally recognized triangle play icon.
- Tappable answer options are rendered as large, clearly bordered buttons with generous spacing, so they visibly invite a tap rather than resembling plain text.
- The QR scan entry point uses a camera viewfinder style icon, matching the visual convention patients already recognize from Mobile Money and other everyday Ghanaian mobile apps.

5. Non Functional Requirements

NFR 1, Performance: end to end processing from speech capture to sign video display should complete within 5 seconds for an average sentence under typical mobile network conditions.

NFR 2, Accessibility: every interactive element must satisfy at least two of the five principles in Section 4 before being considered complete, Feedback and Affordance at minimum.

NFR 3, Haptic reliability: vibration feedback must degrade gracefully on devices or browsers without vibration API support, falling back to visual feedback alone rather than failing silently.

NFR 4, Privacy: the full session transcript is never transmitted to a server without explicit patient action, consistent with FR 4.2.

NFR 5, Low bandwidth tolerance: the application is built as a Progressive Web App and must remain usable, including cached core vocabulary, under intermittent or slow mobile connectivity.

NFR 6, Compatibility: the application must function on current versions of Chrome, Safari, and Firefox on both Android and iOS.

6. Appendix, Vibration Pattern Reference

A single, consistent vibration vocabulary should be implemented once and reused everywhere, rather than defined separately per feature.

Event	Vibration Pattern
Tap selection registered, standard	One short pulse
Audio playback to hearing listener begins	Two short pulses
Audio playback to hearing listener ends	One long pulse
Emergency Triage critical alert selected	Three short, sharp pulses
Session transcript saved	One soft, brief pulse

This table should be treated as the single source of truth for vibration behavior across the app, any new feature added later should reuse one of these patterns rather than inventing a new one, per the Consistency principle in Section 4.4.